

PEG MGF

Pegylated Mechano Growth Factor | 2mg x 10 Vials

Muscle Repair & Satellite Cell Activation | Post-Workout SubQ

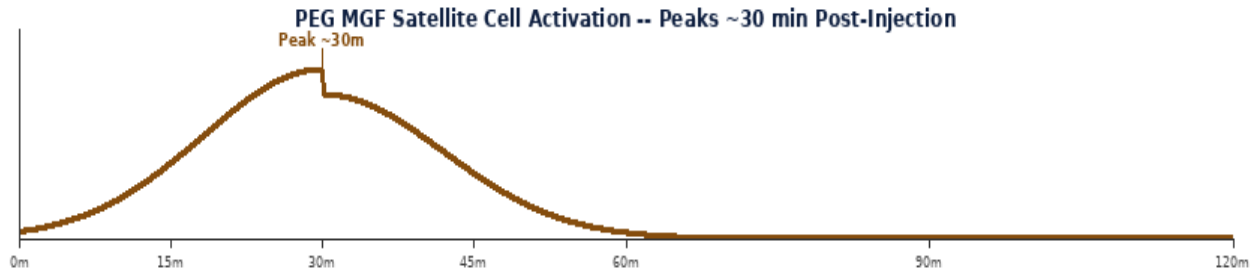
Purity: 99.8% | Endotoxin-Free | Research Use Only

PEG MGF (Pegylated Mechano Growth Factor) is a pegylated splice variant of IGF-1 generated in response to mechanical damage in muscle tissue. The polyethylene glycol (PEG) modification extends its half-life from ~minutes (native MGF) to several days, enabling once or twice weekly dosing. It activates muscle satellite cells and promotes hypertrophy and repair independently of the IGF-1 receptor pathway.



Parameter	Specification
Class	Pegylated IGF-1 splice variant -- mechano growth factor (MGF)
Vial Size	2mg x 10 vials Purity 99.8%
Reconstitution	1 mL BAC Water -> 2 mg/mL (2,000 mcg/mL)
Standard Dose	200-500 mcg per injection 2x per week post-workout
Frequency	2x per week SubQ inject post-workout for best effect
Half-Life	Several days (PEG modification vs ~minutes native MGF)
Mechanism	Activates muscle satellite cells via unique N-terminal peptide domain (not IGF-1R)
Cycle	4-8 weeks on, 4 weeks off short cycles recommended
Storage	2-8 deg C unreconstituted reconstituted: use within 28 days

MECHANISM OF ACTION & RESEARCH BENEFITS



Pathway	Mechanism	Benefit
Satellite Cell Activation	MGF N-terminal domain activates muscle satellite (stem) cells via separate receptor to IGF-1R; satellite cells proliferate and fuse into existing muscle fibres; unique pathway to muscle hyperplasia	Muscle repair acceleration; hyperplasia research; recovery from intense training or injury
PEG Half-Life Extension	Polyethylene glycol modification prevents rapid enzymatic degradation; extends circulation time from minutes to days; allows 2x weekly dosing vs multiple daily injections of native MGF	Practical dosing schedule; sustained satellite cell activation throughout the week
IGF-1 Independence	MGF acts through its own unique C-terminal and N-terminal peptide domains; does NOT raise systemic IGF-1 to the degree of IGF-1 LR3; localised action near site of injection	Muscle-specific effect; lower systemic IGF-1 risk compared to IGF-1 LR3

Benefit	Context
Muscle Satellite Cells	PEG MGF uniquely activates muscle satellite cells -- the muscle stem cells responsible for repair and growth -- via a pathway distinct from IGF-1.
Post-Workout Timing	Inject post-workout near the trained muscle for localised satellite cell activation where muscle damage has occurred.
PEG Convenience	Pegylation extends half-life to days, enabling twice-weekly dosing vs the complex multiple-daily protocol required with native MGF.

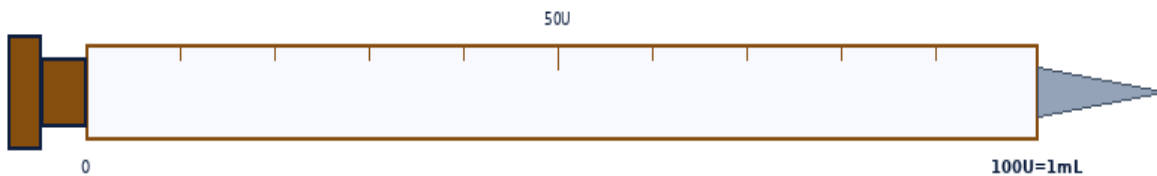
RECONSTITUTION & DOSE CALCULATION

All vials use 1 mL BAC Water. Room temperature. Swirl gently -- never shake.



1 mL BAC Water per vial. Roll gently -- NEVER shake.

Step	Action	Detail
1	Gather	Vial, BAC Water, insulin syringe (29-31G), swabs, sharps.
2	Clean stoppers	Wipe both stoppers. Air-dry 10-15 sec.
3	Draw 1mL BAC	Exactly 1 mL. Expel air bubbles.
4	Inject into vial	Aim at GLASS WALL. Slowly. Never onto powder.
5	Swirl & label	Roll 20-30 sec. Clear within 60 sec. Never shake. Label date. 2-8 deg C. Use within 28 days.



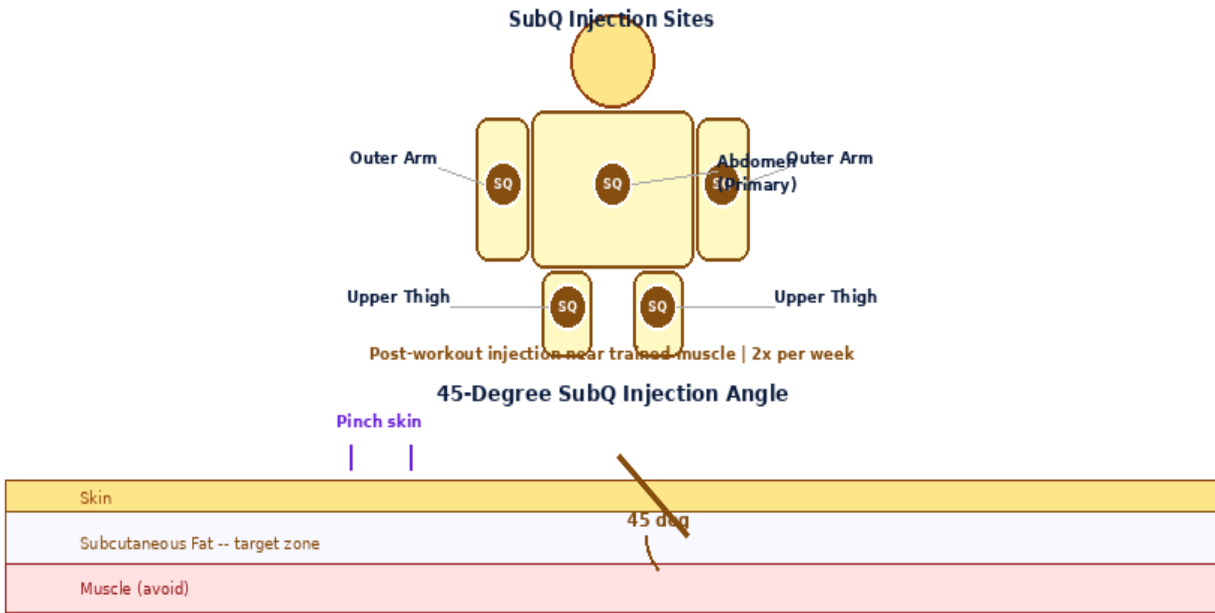
1mL Insulin Syringe | 29-31G | SubQ

Formula: $\text{Vol(mL)} = \frac{\text{Dose(mcg)}}{\text{Conc(mcg/mL)}}$ | **Units:** $\text{Vol} \times 100$

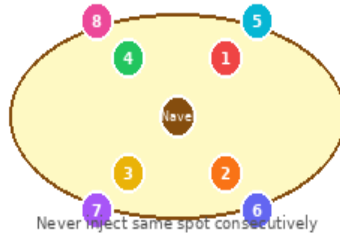
2mg x 10 -- 1mL BAC -> 2.0mg/mL (2,000mcg/mL)

Dose(mcg)	200mcg	300mcg	400mcg	500mcg
Vol(mL)	0.1000	0.1500	0.2000	0.2500
Units	10.0U	15.0U	20.0U	25.0U
Doses/Vial	10	6	5	4

INJECTION TECHNIQUE, PROTOCOLS & GOLDEN RULES



8-Point Rotation Map -- Rotate Every Injection



Step	Action	Detail
1	Draw dose	29-31G insulin syringe. Draw volume per dose table. At 2,000 mcg/mL: 200mcg=0.10mL=10U, 500mcg=0.25mL=25U.
2	Post-workout injection	Inject SubQ within 30-60 min of completing workout, near trained muscle group. Abdomen for upper body; outer thigh for legs.
3	Rotate & dispose	Rotate sites. Press gently. Needle into sharps container.

Protocol	Dose	Frequency	Duration	Notes
Standard	200-300 mcg	2x per week post-workout	4-6wk on 4wk off	Inject near trained muscle group.
Research	400-500 mcg	2x per week post-workout	4-8wk on 4wk off	Higher dose; monitor for side effects.

Side Effect	Likelihood	Management
Injection site reaction	Common	Rotate sites. Use fresh needle. Warm vial to room temp.
Fatigue	Occasional	Post-workout injection context. Usually training-related.
Water retention	Mild	Less than IGF-1 LR3. Monitor.

IGF-1 elevation	Lower than IGF-1 LR3	Contraindicated in cancer history.
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Contraindications: Cancer history, pregnancy, under 18.

1. 1 mL BAC Water -> 2 mg/mL (2,000 mcg/mL). At this concentration: 1U = 20 mcg.
2. Inject post-workout (within 30-60 min) near the trained muscle group.
3. 2x per week dosing is sufficient -- PEG extends half-life to several days.
4. Keep cycles short: 4-8 weeks on, 4 weeks off to prevent desensitisation.
5. No fasting required -- PEG MGF is not dependent on GH/insulin axis.
6. Store 2-8 deg C. Use reconstituted vial within 28 days. Never freeze.
7. Contraindicated in any cancer history -- IGF-1 splice variant stimulates cell growth.
8. Rotate injection sites. Warm vial to room temperature before injecting.

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